

Anticipatory Equity Repricing and the EU Carbon Border Adjustment Mechanism

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ABSTRACT

We ask whether the equities of global steel and aluminium producers repriced by carbon intensity around three 2023 milestones of the European Union's Carbon Border Adjustment Mechanism (CBAM): the European Parliament's adoption of the regulation (18 April), its entry into force (17 May), and the start of the transitional reporting phase (1 October). Using 111 firm-event cumulative abnormal returns for 37 EU and non-EU producers, measured against a leave-one-out benchmark, we estimate a cross-sectional interacted event study of returns on carbon intensity, an EU-domicile indicator, and their interaction, with issuer-clustered inference. The focal coefficient, the carbon slope among non-EU firms, is small and statistically indistinguishable from zero under standardised and log intensity (−0.9 percentage points per standard deviation, 95% confidence interval [−2.6, 0.8], at the [−1, +1] window), but turns positive once a single mis-scaled outlier is neutralised, with the rank-based and deletion specifications statistically significant; the data therefore provide no stable evidence for the hypothesised negative carbon penalty. The EU carbon slope is not separately identified in the raw standardised specification, because within-EU intensity variation is negligible there. Small-sample inference, through CR2 standard errors, a wild cluster bootstrap, and randomization, does not change the pooled picture, and the design is underpowered: the minimum detectable effect is about 1.4 to 2.6 percentage points, and conservative interval-containment equivalence checks cannot rule out effects of one percentage point. We therefore report insufficient evidence to detect a carbon-intensity association in this sample and design, rather than a clean null. A domicile proxy for CBAM exposure and emissions vintages disclosed after the events further caution against a stronger reading.

1. Introduction

The European Union's Carbon Border Adjustment Mechanism (CBAM) extends a carbon price to emissions-intensive imports, with iron and steel and aluminium among the first goods covered ([European Commission, 2023](#)). Its stated purpose is to place imported and domestically produced goods on a comparable carbon-cost footing, so that the bloc's climate ambition is not undercut by production shifting to less regulated jurisdictions. If markets believe the mechanism will raise the cost of carbon-intensive production, the equities of exposed producers should fall in proportion to their carbon intensity once the policy becomes credible. CBAM therefore offers a setting in which to ask whether a major, pre-announced climate regulation is impounded into prices, and whether that repricing tracks the firm characteristic the policy actually targets.

We study three milestones during 2023 that progressively resolved CBAM from proposal to operating regulation: the European Parliament's plenary vote adopting the regulation on 18 April, the entry into force of Regulation (EU) 2023/956 on 17 May (published in the *Official Journal* on 16 May), and the start of the transitional reporting phase on 1 October ([European Commission, 2023](#)). Around each, we ask whether

the equity prices of global steel and aluminium producers moved in proportion to their carbon intensity. The hypothesis of interest is sharpest within the firms the border adjustment is designed to bite: among non-EU firms, higher carbon intensity should be associated with more negative event-window cumulative abnormal returns. Because the emissions vintages underlying the intensity measure were disclosed after the 2023 events for nearly all firms, intensity is a retrospectively measured firm characteristic rather than strictly pre-event information (Section 2.2). This question sits between two literatures. A cross-sectional asset-pricing literature finds that emissions carry a return premium (Bolton and Kacperczyk, 2021, 2023), though one that is contested and sensitive to how emissions are measured and scaled (Aswani et al., 2024a; Sankar et al., 2024); and an event-study literature finds that carbon-regulation news moves carbon-intensive firms, but with effects that vary across the regulatory cycle (Jong et al., 2014; Hengge et al., 2023; Lynch and O'Hagan-Luff, 2024).

We compute one cumulative abnormal return per firm-event against a leave-one-out benchmark, which removes the mechanical self-inclusion that arises when a firm sits inside its own benchmark, and we estimate a cross-sectional interacted event study, not a difference-in-differences model, of firm-event returns on carbon intensity, an EU-domicile indicator, and their interaction, with standard errors clustered by issuer. We report the carbon slope separately for non-EU firms (β_2 , the focal coefficient) and for EU firms ($\beta_2 + \beta_3$), together with their difference (β_3). The sample comprises 111 firm-event cumulative abnormal returns for 37 firms, split 51 EU and 60 non-EU firm-events. At the $[-1, +1]$ window the focal non-EU carbon slope is -0.9 percentage points per standard deviation of intensity, with a 95% confidence interval of $[-2.6, 0.8]$ and $p = 0.30$; the EU carbon slope is not separately identified in the raw standardised specification because within-EU intensity variation is negligible there. The focal slope is not stable: it is insignificant under standardised and log intensity but turns positive once a single mis-scaled firm is neutralised by ranking or deletion, with both outlier-resistant estimates statistically significant and of the opposite sign to the carbon-penalty hypothesis. Small-sample inference through CR2 standard errors, a wild cluster bootstrap, and randomization leaves the pooled focal slope insignificant throughout.

We read this as insufficient evidence to detect a carbon-intensity association in this sample and design, rather than a tightly bounded null. A cross-section of 37 firms observed over three common shocks is thin: the minimum detectable effect at 80% power is about 1.4 to 2.6 percentage points per standard deviation, and equivalence checks cannot rule out effects as small as one percentage point, so wide confidence intervals, not a tight bound around zero, are what the data deliver. Two measurement limitations reinforce the caution. Treatment is proxied by EU versus non-EU domicile rather than by firm-level CBAM exposure; misclassification of this kind is likely to weaken interpretability and may attenuate the association under classical-error conditions, although the direction of bias is not guaranteed. In addition, the carbon-intensity measure uses emissions vintages disclosed after the 2023 events for nearly all firms, so it is not strictly pre-event information. To our knowledge this is the first event-study evidence on anticipatory CBAM pricing in the *equities* of global metals producers; the closest prior study examines Chinese commodity futures rather

than equities (Lin and Zhao, 2023). We also assemble a reusable EU-versus-non-EU panel that links daily prices to firm Scope 1 and Scope 2 carbon intensity, on which the question can be revisited with sharper exposure data.

The remainder of the paper proceeds as follows. Section 1.1 positions the study within the carbon-premium and climate-policy event-study literatures. Section 2 describes the panel, and Section 3 sets out the abnormal-return construction and estimating equations. Section 4 reports the headline and focal results, Section 5 subjects them to robustness lenses, and Sections 6 and 7 interpret the null and conclude.

1.1. Related Literature

Three strands of work locate this paper and, taken together, explain why an imprecisely estimated null around the 2023 CBAM milestones is an informative result rather than an anomalous one.

1.1.1. Carbon risk and the carbon premium

A first literature asks whether carbon emissions are priced in the cross-section of equity returns. Bolton and Kacperczyk (2021) document a carbon premium in US stocks: firms with higher total emissions, and higher year-on-year changes in emissions, earn higher returns after controlling for size, book-to-market, and standard return predictors. Bolton and Kacperczyk (2023) extend the result to a global cross-section of 14,400 firms across 77 countries, reporting higher returns on higher emission levels and growth rates in most countries and all sectors. Read at face value, this evidence implies that a regulatory shock raising the cost of carbon-intensive production should move exposed firms in proportion to their carbon intensity, which is the hypothesis we test for CBAM.

The premium is, however, contested in ways that bear directly on our design. Aswani et al. (2024a) show that the association holds only for vendor-estimated unscaled emissions, not for firm-disclosed emissions, and not at all for emissions intensity scaled by sales; because vendor estimates are highly correlated with financial fundamentals, the measured premium may capture a fundamentals–returns link rather than a distinct carbon factor. The exchange between Bolton and Kacperczyk (2024) and Aswani et al. (2024b) leaves the question open, with the disclosed-emissions result shown to be sensitive to specification. Atilgan et al. (2023) argue that part of the premium reflects mispricing revealed at earnings announcements rather than compensation for risk, and Sankar et al. (2024) find a significant premium for Scope 1 (direct) but not Scope 2 (purchased) emissions among S&P 500 firms. A signal this sensitive to measurement, scaling, and sample is one that need not surface around a single transitional policy event, which frames our null as consistent with, rather than contrary to, the asset-pricing evidence.

1.1.2. Climate-policy and EU-ETS event studies

A second literature studies how equity prices respond to carbon-regulation news, and it supplies both the method we adopt and the contingency we expect. Jong et al. (2014) use the April 2006 publication of first-compliance data, which disclosed an over-allocation of allowances and a sharp fall in the allowance price, to show that the EU Emissions Trading System “bites,” with the share-price impact governed by firms’

carbon intensity. [Hengge et al. \(2023\)](#) extract the surprise component of regulatory actions for more than 2,000 European firms over 2011–2021 and find that carbon-price-raising events generate negative abnormal returns that grow with emission intensity. The effect is not constant, however: [Lynch and O’Hagan-Luff \(2024\)](#) report that positive environmental news raised returns only in the later years of EU ETS Phase 3 (2018–2020) and had an insignificant effect in the earlier years (2013–2017), which is the kind of transitional, low-salience window that CBAM’s 2023 events occupy. Evidence on the real-side channel is similarly muted: [Branger et al. \(2016\)](#) find no significant effect of the carbon price on net imports of steel and cement, and hence no detectable short-run carbon leakage in the very sectors CBAM targets. Our design follows the standard event-study apparatus surveyed by [MacKinlay \(1997\)](#) and [Kothari and Warner \(2007\)](#), whose treatment of short-horizon specification and statistical power also bears on how a small-sample null should be read.

1.1.3. CBAM and the anticipation of regulatory shocks

The third strand concerns CBAM itself and whether a pre-announced regulation is impounded in advance. The mechanism covers cement, iron and steel, aluminium, fertilisers, electricity, and hydrogen. Its transitional phase, which began in October 2023, imposed only reporting obligations, with no certificate cost; the definitive regime, under which importers surrender CBAM certificates, entered into force on 1 January 2026 ([European Commission, 2023](#)). During the 2023 windows we study, then, the near-term cash cost to exposed producers was small, which is a primary-source reason to expect either anticipatory pricing during the long 2021–2023 legislative process or a muted reaction at the 2023 milestones themselves. Direct market-reaction evidence remains thin. The closest study is [Lin and Zhao \(2023\)](#), who examine *Chinese commodity futures* and find a significant decline in steel-rebar and aluminium futures returns around the CBAM announcement, with a stronger effect for aluminium than for steel. That setting differs from ours in instrument (futures versus equities), market (China versus a global cross-section of producers), and horizon, so it neither anticipates nor contradicts an equity null. No published event study, to our knowledge, examines whether CBAM’s 2023 regulatory milestones were priced into the *equities* of global steel and aluminium producers in proportion to carbon intensity. Our paper fills that gap. Across a cross-sectional interacted event study with issuer-clustered inference, a focal within-non-EU test, transformation and influence checks, and small-sample inference, we find no stable evidence of the hypothesised negative carbon-intensity association, and we quantify the power the design can bring to bear rather than asserting that no effect exists. The transitional-phase leniency, the contested and measurement-sensitive nature of the carbon premium, the documented time-variation in EU ETS valuation effects, and the limited power of a small cross-section together make this null economically interpretable rather than a mere failure to reject.

2. Data

The study combines daily equity prices with firm-level carbon and revenue data for a panel of steel and aluminium producers. This section describes the sample and

Table 1

CBAM event chronology. Each 21-session price window is centred on the official milestone; day 0 is the first trading session on or after it.

Event	Milestone date	Day 0 (session)	Official action
E1	2023-04-18	2023-04-18	European Parliament plenary vote adopting the CBAM Regulation
E2	2023-05-17	2023-05-17	Entry into force of Regulation (EU) 2023/956 (published OJ L130, 16 May 2023)
E3	2023-10-01	2023-10-02	Start of the CBAM transitional reporting phase (rules in Implementing Reg. (EU) 2023/1773)

Notes. Sources: European Parliament press release (18 Apr 2023); Regulation (EU) 2023/956, OJ L130 (16 May 2023, in force 17 May 2023); Implementing Regulation (EU) 2023/1773 (17 Aug 2023), which governs the transitional reporting phase that began 1 Oct 2023. Windows are disjoint.

the event chronology, the construction and audit of carbon intensity, the coverage of the merge, and the descriptive statistics. Two features of the data that bear on interpretation, an EU-domicile proxy for CBAM exposure and emissions vintages disclosed after the events, are flagged here and taken up in Section 6.

2.1. Sample and event chronology

The price file covers 37 listed steel and aluminium producers across the European Union and the rest of the world, with daily adjusted closes around three CBAM milestones in 2023. Table 1 gives the chronology. Each window is centred on the official milestone and spans roughly ten trading sessions on either side; day 0 is the first trading session on or after the milestone. The three milestones are the European Parliament's plenary vote adopting the CBAM Regulation (18 April 2023), the entry into force of Regulation (EU) 2023/956 (17 May 2023, after publication in the *Official Journal* on 16 May), and the start of the transitional reporting phase (1 October 2023, first traded on 2 October). The windows are disjoint, so no return is reused across events. After computing returns and leave-one-out abnormal returns (Section 3), the headline cross-section holds 111 firm-event cumulative abnormal returns, 51 for EU producers and 60 for non-EU producers; the narrow windows hold 109 to 111, the small shortfall reflecting a few local-market non-trading sessions. The event set comprises the three official 2023 milestones of the CBAM legislative calendar (Table 1); extending the analysis to the December 2022 political agreement or the August 2023 implementing regulation would require price data outside the assembled windows and is left to future work.

2.2. Carbon intensity and its audit

Firm carbon intensity is the sum of Scope 1 and Scope 2 emissions divided by revenue converted to U.S. dollars with a fixed 2023 exchange-rate map for the reporting currencies. The cross-section is strongly right-skewed, with a mean of 0.19 and a median of 0.0006 tonnes of CO₂-equivalent per dollar of revenue, a maximum of 7.03, and a standard deviation of 1.14. An audit of the extremes (Table ??) traces the maximum to a single firm, JSW Steel, whose reported revenue of about \$21,000 after currency conversion is roughly six orders of magnitude too small for a major producer. This is an invalid-denominator unit error rather than a real intensity, and it is the sole driver of both the maximum and the large mean-median gap. We report the raw construction and then, in Section 5, standardise, log, rank, winsorise, and delete this observation to show how the focal estimate depends on it. A further eight firms report zero or

Table 2

Descriptive statistics for the headline estimation sample (111 firm-events, 37 firms). Cumulative abnormal returns are in percentage points.

Variable	<i>N</i>	Mean	SD	Min	Median	Max
CAR [0, +1] (pp)	109	-0.00	2.21	-8.71	0.18	6.49
CAR [-1, +1] (pp)	110	-0.00	2.66	-9.64	0.10	8.68
CAR [0, +3] (pp)	110	0.00	3.29	-10.91	0.36	6.91
CAR full window (pp)	111	-0.00	4.66	-14.99	0.35	13.83
Carbon intensity (t/\$)	111	0.1905	1.1446	0.0000	0.0006	7.0273
Carbon intensity (z)	111	-0.00	1.00	-0.17	-0.17	6.00
Revenue (\$bn)	111	21.1	20.8	0.000	12.3	78.1
EU domicile (0/1)	111	0.46	0.50	0.00	0.00	1.00

Notes. Cumulative abnormal returns use the leave-one-out benchmark; by construction their cross-sectional mean is approximately zero at each window. Carbon intensity is (Scope 1 + Scope 2)/Revenue^{USD}; its z-score is used in the regressions.

missing Scope 2 emissions, coded as zero; genuine zero purchased-energy emissions are implausible for metals producers, so these are best read as missing.

Two limitations of the intensity measure matter for identification. First, corporate Scope 1 plus Scope 2 intensity is a coarse proxy for CBAM liability, which concerns embedded emissions in specific covered goods exported to the EU rather than whole-firm emissions. Second, the emissions vintages are not strictly pre-event: 32 of 37 firms report fiscal-year 2023 figures and four report 2024, both disclosed after the 2023 event windows, so only one firm's vintage clearly predates the events. The measure therefore carries look-ahead information for nearly all firms, which we cannot remove without point-in-time 2022 emissions.

2.3. Coverage, matching, and the domicile proxy

The price and emissions files are merged on the listing symbol. One issuer, POSCO, is listed in the emissions file under two symbols, the local Korean listing and its U.S. depositary receipt, but only the depositary receipt appears in the price file, so POSCO enters the panel exactly once and no issuer is double counted. All 37 price tickers receive a carbon intensity. Treatment is defined by EU versus non-EU domicile. This is a proxy, not a measure of CBAM exposure: a non-EU firm with few EU sales has little direct exposure, while an EU firm may import covered goods or compete with affected imports. Misclassification of this kind is likely to weaken interpretability and may attenuate any true association under classical-error conditions, although the direction of bias is not guaranteed; either way it limits what a null can establish. We return to this in Section 6 and list the firms and their classification in Table ??.

2.4. Descriptive statistics

Table 2 summarises the estimation sample. Cumulative abnormal returns are small and centred near zero at every window, with standard deviations of two to five percentage points. Their cross-sectional mean is approximately zero by construction, because the leave-one-out benchmark demeans returns within each date (Section 3). Revenue spans a wide range, from small specialty producers to large integrated mills, which motivates the log-revenue control in the regressions.

3. Methodology

We study three CBAM milestones during 2023 and ask whether the cross-section of event-window returns lines up with firms' carbon intensity. The design is a cross-sectional interacted event study (MacKinlay, 1997; Kothari and Warner, 2007) built on firm-event cumulative abnormal returns, with an EU-domicile indicator, carbon intensity, and their interaction as regressors. It is not a difference-in-differences model: there is no separately observed pre- and post-treatment outcome and no treatment-by-post interaction, so we avoid that label and the causal language that accompanies it. This section defines the return measure, the benchmark, the outcome, the estimating equation and the signs it predicts, and the inference and power procedures. The equations below correspond to the estimators used in the empirical analysis.

3.1. Returns and the leave-one-out benchmark

For firm i on trading day t we compute the simple return from the adjusted close,

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1, \quad (1)$$

within each ticker-window block, so returns never span across two event windows. Returns are winsorized at the first and ninety-ninth percentiles to limit the influence of a few extreme observations.

The abnormal return uses a leave-one-out benchmark equal to the equal-weighted mean return of all other firms trading on the same day,

$$\tilde{r}_{i,t} = \frac{1}{N_t - 1} \left(\sum_{j \in S_t} r_{j,t} - r_{i,t} \right), \quad AR_{i,t} = r_{i,t} - \tilde{r}_{i,t}, \quad (2)$$

where S_t is the set of firms trading on day t and $N_t = |S_t|$. This benchmark removes a firm's own return from its comparison group, but it is a within-sample construction rather than an externally estimated expected return, and it carries an algebraic property. Writing $\bar{r}_t$ for the full cross-sectional mean,

$$AR_{i,t} = \frac{N_t}{N_t - 1} (r_{i,t} - \bar{r}_t), \quad (3)$$

so the abnormal return is a scaled cross-sectional demeaning of the same-day sample. Three consequences follow. First, the abnormal returns sum to (approximately) zero across firms each day, so the EU and non-EU average paths in Figure 2 offset each other mechanically and their visual separation is not independent evidence. Second, the benchmark still contains firms exposed to the same policy news, so it does not purge common treatment. Third, it induces cross-sectional dependence that firm clustering alone does not fully address. Identification of any carbon effect therefore rests on the within-non-EU slope and the between-group interaction, not on the level of the benchmark; a genuinely external market or sector benchmark, unavailable in the assembled data, is noted as a limitation in Section 6.

3.2. Cumulative abnormal returns and event windows

The outcome is the cumulative abnormal return over a narrow window around day 0,

$$CAR_{i,e}^{(h)} = \sum_{t \in h} AR_{i,t}, \quad (4)$$

one observation per firm-event pair, where i indexes firms, e indexes the three CBAM events, and h indexes the event-window specification. We use three short, non-overlapping windows as the primary specifications, $[0, +1]$, $[-1, +1]$, and $[0, +3]$ trading days, which limit contamination from unrelated macro and firm news, and retain the full ± 10 -day window only as a sensitivity check. Because the three milestones are heterogeneous, we report each event separately as well as pooled. Carbon intensity is the sum of Scope 1 and Scope 2 emissions scaled by revenue in U.S. dollars,

$$CI_i = \frac{\text{Scope1}_i + \text{Scope2}_i}{\text{Revenue}_i^{\text{USD}}}, \quad (5)$$

standardized across the firm-event cross-section to CI_i^{std} so its coefficients are read per standard deviation. The emissions figures entering CI_i are fiscal-year vintages disclosed after the 2023 events for nearly all firms (Section 2.2), so CI_i is a retrospectively measured firm characteristic rather than pre-event information.

3.3. Headline specification and predicted signs

The headline estimate is a cross-sectional regression of firm-event CARs on the EU indicator, carbon intensity, and their interaction,

$$CAR_{i,e}^{(h)} = \alpha + \beta_1 EU_i + \beta_2 CI_i^{\text{std}} + \beta_3 (EU_i \times CI_i^{\text{std}}) + \beta_4 \log \text{Revenue}_i^{\text{USD}} + \delta_e + \varepsilon_{i,e}^{(h)}, \quad (6)$$

where EU_i indicates an EU producer, δ_e are event fixed effects, and $\log \text{Revenue}$ controls for size. The coefficient mapping matters for which hypothesis each term tests. Here β_2 is the carbon slope for non-EU firms, $\beta_2 + \beta_3$ is the carbon slope for EU firms, and β_3 is the difference between the two. A simple incidence argument fixes the predicted signs. If CBAM raises the expected cost of carbon-intensive production for producers exporting into the EU, then higher-carbon non-EU firms should fall more around credible CBAM news, so $\beta_2 < 0$. EU producers, partly shielded by the border adjustment and by continued free allocation, should react less, so the EU slope $\beta_2 + \beta_3$ should be closer to zero and the difference $\beta_3 = (\beta_2 + \beta_3) - \beta_2$ should be positive. The focal test is therefore β_2 , not β_3 ; we estimate (6) in a form that returns β_2 and $\beta_2 + \beta_3$ directly, each with a confidence interval. Because each firm appears in up to three event windows, standard errors are clustered by issuer (Cameron et al., 2008).

3.4. Focal within-non-EU test

The hypothesis is stated for non-EU firms, whose competitiveness CBAM directly affects. We therefore also estimate the carbon slope inside the non-EU group alone,

$$CAR_{i,e}^{(h)} = \gamma_0 + \gamma_1 CI_i^{\text{std}} + \delta_e + \eta_{i,e}^{(h)}, \quad i \in \text{non-EU}, \quad (7)$$

with issuer-clustered standard errors. A negative and significant γ_1 , which equals β_2 in (6) up to the controls, would be the cleanest evidence for a carbon-border penalty.

Table 3

Pooled cross-sectional interacted event study of firm-event cumulative abnormal returns on carbon intensity, EU domicile, and their interaction, by event window. Coefficients in percentage points with 95% confidence intervals in brackets; issuer-clustered standard errors; event fixed effects and log-revenue included.

	[0, +1]	[-1, +1]	[0, +3]
β_2 non-EU carbon slope (pp)	-0.13 [-1.10, 0.84]	-0.90 [-2.60, 0.80]	0.32 [-1.52, 2.16]
$\beta_2 + \beta_3$ EU carbon slope (pp)	246.02 [-933.22, 1425.25]	810.70 [-879.22, 2500.63]	143.98 [-2159.64, 2447.60]
β_3 interaction (pp)	246.14 [-933.47, 1425.76]	811.60 [-879.05, 2502.26]	143.66 [-2161.07, 2448.39]
<i>N</i> firm-events	109	110	110

Notes. The model is $CAR_{i,e}^{(h)} = \alpha + \beta_1 EU_i + \beta_2 CI_i^z + \beta_3 (EU_i \times CI_i^z) + \beta_4 \log Rev_i + \delta_e + \varepsilon_{i,e}^{(h)}$, reparameterised so β_2 is the non-EU carbon slope, $\beta_2 + \beta_3$ the EU carbon slope, and β_3 their difference. CI^z is standardised carbon intensity, so slopes are per one standard deviation. The EU carbon slope is not identified in this raw standardised specification: within-EU intensity variation is near zero (Table 5), which inflates $\beta_2 + \beta_3$ and β_3 and their intervals.

3.5. Inference and power

With only three common event dates, conventional cluster-robust inference can be optimistic, so we report several alternatives for each carbon slope. Alongside issuer-clustered CR1 standard errors we compute the bias-reduced CR2 correction, a wild cluster bootstrap with Rademacher weights at the issuer level (9,999 draws), and randomization inference (10,000 draws) that permutes the EU label across firms, for the EU slope and the interaction, or intensity within event, for the non-EU slope. These issuer-level procedures address the repetition of firms across up to three event windows; they do not by themselves address dependence across firms exposed to the same three policy shocks, which the leave-one-out benchmark induces (Section 3.1), so the effective number of independent shocks remains three. Because a failure to reject zero is not evidence of no effect, we also quantify power. We report the minimum detectable effect at 80% power and 5% size for each slope, and an equivalence check that asks whether the 95% confidence interval for each slope lies entirely inside ± 1 and ± 2 percentage points per standard deviation of intensity. Requiring the full 95% interval to fit inside the margin is a stricter criterion than a conventional two one-sided test at the 5% level, which would use a 90% interval, so the check is conservative rather than a standard TOST. All estimates and intervals are reported in percentage points.

4. Empirical Results

The sample is a cross-section of 111 firm-event cumulative abnormal returns for 37 firms, 51 EU and 60 non-EU firm-events, across the three 2023 CBAM milestones. The results point in one direction: event-window returns do not line up robustly with carbon intensity, in the pooled cross-section or within the non-EU group that CBAM most directly affects, and the design is too thin to rule out economically meaningful effects.

4.1. Headline cross-sectional event study

Table 3 reports the estimate of (6) at each narrow window, with issuer-clustered standard errors and coefficients in percentage points. The focal coefficient is the non-EU carbon slope β_2 . It is small and statistically indistinguishable from zero at

Table 4

Event-by-event and pooled estimates of the carbon slopes at the $[-1, +1]$ window. Percentage points; 95% confidence intervals in brackets; issuer-clustered.

Event	β_2 non-EU slope (pp)	p	$\beta_2 + \beta_3$ EU slope (pp)	N
E1 (EP vote)	-1.12 [-3.48, 1.25]	0.355	1228.54 [-2155.94, 4613.02]	37
E2 (in force)	-0.30 [-3.48, 2.89]	0.855	1443.35 [-1630.06, 4516.76]	37
E3 (transitional)	-1.30 [-2.57, -0.02]	0.046	-233.85 [-2981.43, 2513.73]	36
Pooled	-0.90 [-2.60, 0.80]	0.300	810.70 [-879.22, 2500.63]	110

Notes. β_2 is the focal coefficient (non-EU firms). The EU slope is not separately identified in the raw standardised specification (Table 5).

every window: -0.13 percentage points per standard deviation at $[0, +1]$ (95% CI $[-1.10, 0.84]$), -0.90 at $[-1, +1]$ ($[-2.60, 0.80]$), and $+0.32$ at $[0, +3]$ ($[-1.52, 2.16]$), with p -values of 0.80, 0.30, and 0.73. The EU carbon slope $\beta_2 + \beta_3$ and the interaction β_3 carry point estimates in the hundreds of percentage points with intervals that run into the thousands. These are not economic magnitudes; they are the signature of an unidentified coefficient, which we document next. Event-by-event estimates at the $[-1, +1]$ window, in Table 4, sharpen the picture: the non-EU slope is insignificant at the first two milestones and when pooled, while the E3 (transitional-phase) estimate of -1.30 percentage points is marginally significant under conventional issuer-clustered inference ($p = 0.046$). Because multiple event-specific tests are conducted and no event-specific small-sample adjustment is reported, we treat the E3 result as isolated rather than robust. Figure 1 shows the same event-by-event picture graphically, as mean $[-1, +1]$ CARs by domicile with 95% bands: the EU and non-EU means straddle zero at every milestone and no consistent gap opens.

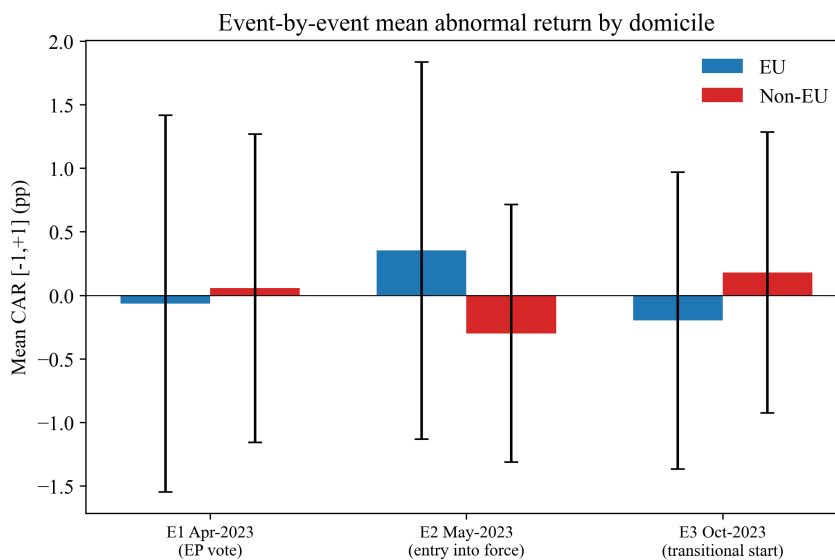


Figure 1: Mean cumulative abnormal return over $[-1, +1]$ by event and domicile, with 95% confidence bands. Group means use the leave-one-out benchmark, so by the demeaning identity in (3) the EU and non-EU means are algebraically linked and their separation is not independent evidence of a differential reaction.

Table 5

Identification diagnostics for the pooled $[-1, +1]$ specification. Near-zero within-EU intensity variation makes the EU carbon slope collinear with the EU indicator.

Diagnostic	Value
SD of carbon intensity, EU firms	0.0004
SD of carbon intensity, non-EU firms	1.5314
Design-matrix condition number	718
VIF, EU indicator	127,929
VIF, EU $\times$ carbon (EU slope)	128,054
VIF, non-EU carbon slope	7.51
VIF, log revenue	7.73

Notes. Variance-inflation factors above 10^4 on the EU indicator and the EU carbon slope confirm the EU slope ($\beta_2 + \beta_3$) is not separately identified in the raw standardised specification; the non-EU slope (β_2) is well conditioned (VIF < 8).

4.2. Why the EU slope is not identified

Table 5 explains the large EU-slope estimates. Within-EU carbon intensity has a standard deviation of 0.0004, against 1.53 for non-EU firms: EU producers are effectively a single point in intensity. The EU-interacted intensity term is therefore almost perfectly collinear with the EU indicator, with variance-inflation factors above 10^4 for both and a design-matrix condition number of 718, so $\beta_2 + \beta_3$ and β_3 cannot be separated from the EU level and their intervals explode. The non-EU slope, by contrast, is well conditioned (VIF below 8). This resolves an apparent tension between the two carbon estimates. A large negative EU-by-carbon interaction and a positive within-non-EU slope are not two views of one effect: the first is an unidentified between-group difference and the second is a separately identified non-EU slope. Only the non-EU slope is interpretable, and it is not distinguishable from zero. This non-identification is a property of the raw standardised specification, in which the erroneous non-EU extreme compresses the EU intensity range to a point; under the log, rank, and deletion treatments of Section 5 the EU slope takes ordinary magnitudes with wide intervals (Table 8), although within-EU variation remains limited.

4.3. Small-sample inference

Because the effective number of policy shocks is three, we do not rely on a single standard-error formula. Table 6 reports, for each carbon slope and window, p -values from issuer-clustered CR1 and CR2 standard errors, a wild cluster bootstrap, and randomization inference. For the pooled focal non-EU slope the four methods agree: it is insignificant at every window, with p -values from 0.11 to 0.81. The wild bootstrap and randomization do not overturn the clustered result. All four procedures operate at the issuer level, however, so they address the repetition of firms across windows rather than dependence across firms exposed to the same three policy shocks (Section 3.5). The unidentified EU slope and interaction are likewise insignificant, as their inflated standard errors require.

Table 6

Small-sample inference for the carbon slopes. p -values from issuer-clustered CR1 and CR2 standard errors, a wild cluster bootstrap (Rademacher, 9,999 draws), and randomization inference (10,000 draws).

Coefficient	Est. (pp)	p_{CR1}	p_{CR2}	p_{wild}	p_{RI}
<i>Window [0, +1]</i>					
β_2 non-EU slope	-0.13	0.797	0.816	0.809	0.677
$\beta_2 + \beta_3$ EU slope	246.02	0.683	0.693	0.690	0.400
β_3 interaction	246.14	0.683	0.693	0.690	0.736
<i>Window [-1, +1]</i>					
β_2 non-EU slope	-0.90	0.300	0.407	0.413	0.113
$\beta_2 + \beta_3$ EU slope	810.70	0.347	0.431	0.323	0.430
β_3 interaction	811.60	0.347	0.431	0.322	0.807
<i>Window [0, +3]</i>					
β_2 non-EU slope	0.32	0.734	0.755	0.753	0.524
$\beta_2 + \beta_3$ EU slope	143.98	0.903	0.906	0.909	0.485
β_3 interaction	143.66	0.903	0.907	0.909	0.875

Notes. Randomization permutes the EU label across firms (for the EU slope and interaction) or intensity within event (for the non-EU slope); 37 issuer clusters. The focal non-EU slope is insignificant under every method.

Table 7

Power and equivalence for the carbon slopes. Minimum detectable effect at 80% power / 5% size, and equivalence checks requiring the 95% confidence interval to lie inside ± 1 and ± 2 percentage points per standard deviation of intensity (a stricter criterion than a conventional 5% two one-sided test, which would use a 90% interval).

Window	Coefficient	Estimate (pp) [95% CI]	MDE (pp)	Equiv. ± 1 pp	Equiv. ± 2 pp
[0, +1]	β_2 non-EU	-0.13 [-1.10, 0.84]	1.38	NO	YES
[0, +1]	$\beta_2 + \beta_3$ EU	246.02 [-933.22, 1425.25]	1685.61	NO	NO
[-1, +1]	β_2 non-EU	-0.90 [-2.60, 0.80]	2.43	NO	NO
[-1, +1]	$\beta_2 + \beta_3$ EU	810.70 [-879.22, 2500.63]	2415.59	NO	NO
[0, +3]	β_2 non-EU	0.32 [-1.52, 2.16]	2.63	NO	NO
[0, +3]	$\beta_2 + \beta_3$ EU	143.98 [-2159.64, 2447.60]	3292.81	NO	NO

Notes. Equivalence is declared only when the 95% interval lies inside the bound, a conservative interval-containment criterion rather than a standard TOST. It fails at ± 1 pp for every coefficient and holds at ± 2 pp only for the non-EU slope at [0, +1]; the EU-slope bounds run to thousands of points. The design cannot rule out carbon effects up to roughly one to two-and-a-half percentage points.

4.4. Power and equivalence

A null is only as strong as the effects it can exclude. Table 7 reports the minimum detectable effect and the interval-containment equivalence checks (Section 3.5). At 80% power the design can detect a non-EU carbon slope of about 1.4 percentage points per standard deviation at [0, +1], rising to 2.6 at [0, +3]; effects smaller than this are simply beyond its resolution. The equivalence checks confirm the point: the 95% interval never fits inside ± 1 percentage point, and fits inside ± 2 points only for the non-EU slope at [0, +1]. The EU-slope bounds are meaningless, running to thousands of points. We cannot rule out carbon-intensity effects of one to two and a half percentage points per standard deviation, so the data deliver wide confidence intervals rather than a tight bound around zero.

4.5. Event-study visualization

Figure 2 plots the average cumulative abnormal return for EU and non-EU firms across relative trading days, with 95% bands. The two paths move together around

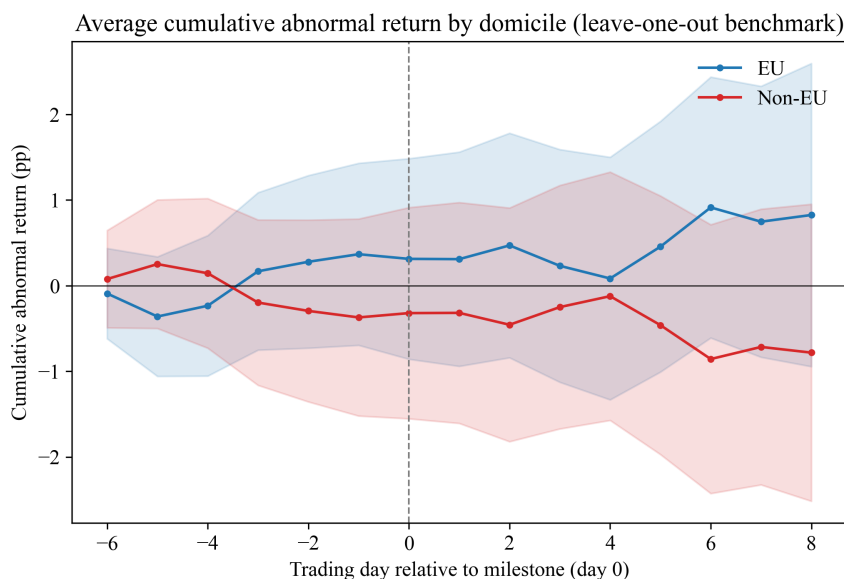


Figure 2: Average cumulative abnormal return for EU and non-EU steel and aluminium producers by trading day relative to the milestone (day 0), pooled over the three 2023 CBAM events, with 95% bands. Abnormal returns use the leave-one-out benchmark, so by the demeaning identity in (3) the EU and non-EU average paths offset each other mechanically; their separation is not independent evidence of a differential reaction and does not track carbon intensity.

each announcement and drift apart over the following sessions, but the separation is small relative to its confidence band, is a mechanical consequence of the shared benchmark, and does not line up with carbon intensity in the regressions.

4.6. Interpretation

The leave-one-out benchmark removes direct self-inclusion but retains the mechanical and dependence properties of a within-sample cross-sectional mean. Under this design and several small-sample inference methods the pooled focal carbon coefficient remains indistinguishable from zero, the between-group slope is unidentified in the raw standardised specification, and the power to detect a modest effect is limited. The reading is narrow: we find no stable evidence of the hypothesised negative carbon-intensity association in event-window returns around the three 2023 CBAM milestones, and the design cannot rule out effects up to a couple of percentage points. Section 5 shows that the two outlier-resistant specifications that produce significant coefficients do so with positive sign, and that both depend on the treatment of a single mis-scaled firm.

5. Robustness and Additional Tests

We probe the focal non-EU slope where it is most fragile: the treatment of carbon intensity, the influence of individual firms, and the vintage of the emissions data. The

Table 8

Carbon-intensity transformations, pooled $[-1, +1]$ window. The focal non-EU slope is insignificant under standardised and log intensity but turns positive, and statistically significant, under rank and after deleting the single invalid-denominator firm.

Transformation	β_2 non-EU (pp)	p	$\beta_2 + \beta_3$ EU (pp)	p
Standardised (raw)	-0.90 [-2.60, 0.80]	0.300	810.70 [-879.22, 2500.63]	0.347
Log intensity	0.41 [-1.15, 1.97]	0.606	0.15 [-0.45, 0.76]	0.620
Rank	0.99 [0.15, 1.84]	0.022	0.21 [-0.55, 0.97]	0.587
Winsorised 1/99	-0.90 [-2.60, 0.80]	0.300	810.70 [-879.22, 2500.63]	0.347
Drop invalid denominator	0.91 [0.22, 1.61]	0.010	0.23 [-0.42, 0.89]	0.489
Drop vintage ≥ 2023 (infeasible)	$n = 1$ firm (FY2022 only); not estimable			

Notes. Winsorising at 1/99 does not clip the lone extreme among 37 firms, so it reproduces the standardised result exactly. The look-ahead subsample (emissions vintage before 2023) leaves one firm and cannot be estimated.

checks show that once the single mis-scaled firm is down-weighted or removed, the estimates turn positive, with the rank-based and deletion specifications statistically significant; no specification supports the hypothesised negative carbon penalty.

5.1. Carbon-intensity transformations

Table 8 re-estimates the pooled $[-1, +1]$ specification under five treatments of intensity. Under standardised and log intensity the non-EU slope is insignificant (-0.90 , $p = 0.30$; $+0.41$, $p = 0.61$). Under ranks it becomes $+0.99$ percentage points ($p = 0.02$), and after deleting the single invalid-denominator firm it becomes $+0.91$ ($p = 0.01$). Both significant estimates are positive, the opposite sign to the carbon-penalty hypothesis, and both arise only once the mis-scaled JSW Steel observation is down-weighted by ranking or removed. Winsorising at the first and ninety-ninth percentiles does not clip a lone extreme among 37 firms, so it reproduces the standardised result exactly. No transformation yields a negative, significant slope of the predicted sign. After the invalid observation is down-weighted or removed, the three relevant estimates (log, rank, and deletion) share a positive direction: the data provide no stable evidence for the hypothesised negative carbon penalty, and the positive corrected-data estimates are an opposite-direction finding that would require explanation and replication, not a null. Figure 3 plots the outcome against standardised intensity by event and region and makes the source of the fragility visible: the non-EU cloud is anchored at low intensity with one far-right point, and the EU firms occupy a single vertical strip, which is why the EU slope cannot be identified.

5.2. Influential firms

Leverage and Cook's distance for the pooled $[-1, +1]$ regression identify JSW Steel as the highest-leverage observation (hat value 0.35), a direct consequence of its erroneous intensity, followed by a handful of EU firms with large event-window moves. Excluding JSW Steel changes the focal slope from -0.90 percentage points ($p = 0.30$) to $+0.91$ ($p = 0.01$), away from the predicted penalty and in line with the rank-based estimate. The raw negative coefficient is therefore the estimate most dependent on the erroneous observation, while the outlier-resistant estimates agree in their positive direction, although not uniformly in significance.

Anticipatory CBAM Repricing

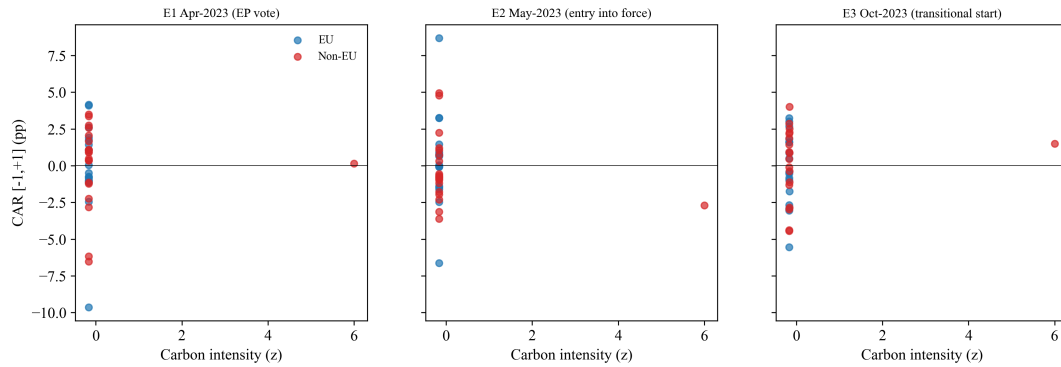


Figure 3: Cumulative abnormal return $[-1, +1]$ against standardised carbon intensity, by event and domicile. The single non-EU point at the far right is JSW Steel, whose intensity is a unit error; EU firms cluster in a narrow intensity strip.

5.3. Emissions vintage

A robustness check would drop firms whose emissions vintage postdates the events, to guard against look-ahead. On these data the guard is not operable: only one firm reports a pre-2023 vintage, so the subsample cannot be estimated. We report the vintage distribution instead (32 firms fiscal-year 2023, four 2024, one 2022) and treat the pervasive look-ahead as a limitation rather than something the current data can correct.

5.4. Exploratory analyses

Three further exercises, a structural-break (backward sup-augmented Dickey-Fuller) sample, a trend-scanning statistic, and a random-forest classifier, were examined and are described in Appendix ???. Each is ill-suited to 21-session firm-event windows, conditions on realized post-event prices, or rests on an extremum statistic, so we do not report their estimates and none informs the results above. The economically motivated checks in this section carry the robustness argument.

6. Discussion

The central result is that event-window returns do not load robustly on carbon intensity around the 2023 CBAM milestones: the focal non-EU slope is insignificant under the primary specifications, and the outlier-resistant transformations yield small positive estimates, two of them statistically significant, opposite in direction to the hypothesised penalty. This section interprets that pattern of no stable evidence for the hypothesised negative association. It introduces no new estimates; it asks why a reaction predicted by a literal reading of the carbon-premium literature need not appear in this setting.

Four explanations are consistent with the evidence, and the data cannot fully separate them. Markets may not have reacted; the selected dates may have carried little new information after a long, well-telegraphed legislative process; the sample may lack the power to detect a modest effect; or the treatment and intensity proxies may be too noisy to reveal one. The first two are economically plausible. The transi-

tional phase that began in October 2023 imposed only reporting obligations, with no certificate cost until the definitive regime took force in January 2026 ([European Commission, 2023](#)), so the 2023 milestones conveyed little immediate cash-flow consequence, and the long 2021–2023 process gave markets ample opportunity to impound whatever cost they anticipated beforehand. The channel is also contested: the cross-sectional carbon premium is sensitive to whether emissions are vendor-estimated or firm-disclosed and to whether they are scaled by sales ([Aswani et al., 2024a](#)), and it appears for direct but not purchased emissions ([Sankar et al., 2024](#)), so a signal this fragile in broad cross-sections need not surface around three dates. The EU-ETS event-study evidence points the same way, with insignificant valuation effects in the earlier, transitional years of Phase 3 ([Lynch and O’Hagan-Luff, 2024](#)) and no detectable short-run carbon leakage in steel and cement ([Branger et al., 2016](#)).

The latter two explanations are design limitations that we cannot dismiss, and they are why we frame the result as insufficient evidence rather than a precise one. Power is binding: with 37 firms and three common shocks, the minimum detectable non-EU slope is one to two and a half percentage points per standard deviation, and equivalence checks cannot exclude effects of one point. Measurement is the second constraint. Treatment is proxied by EU versus non-EU domicile, not by firm-level CBAM exposure, which depends on EU-export share, covered-product share, and embedded emissions; misclassification of this kind is likely to weaken interpretability and may attenuate the association under classical-error conditions, although the direction of bias is not guaranteed. The intensity measure is corporate Scope 1 plus Scope 2 divided by revenue, a coarse stand-in for embedded emissions in covered goods, and its vintages postdate the events for nearly all firms, so it is not strictly pre-event information. Finally, the leave-one-out benchmark is a within-sample construction whose demeaning identity forces the group paths to offset, and the EU carbon slope is not separately identified in the raw standardised specification because EU firms show almost no raw intensity variation.

Read through the lens of market efficiency, then, our evidence says only that we do not detect anticipatory repricing of metals equities by carbon intensity around these specific announcements, with this proxy and this benchmark, at the power the sample affords. It is consistent with prices having already reflected the policy, with the transitional phase being judged immaterial in the near term, and with an effect too small for the design to resolve; it is not evidence that carbon risk is unpriced in general, and it coexists with the asset-pricing evidence that it sometimes is ([Bolton and Kacperczyk, 2021, 2023](#)).

These limits also map the path to a sharper test. A firm-level exposure measure built from EU-export share, covered-product share, and embedded emissions would replace the domicile proxy; point-in-time emissions vintages would remove the look-ahead; an externally traded market or sector benchmark estimated over a pre-event window would replace the within-sample mean; and a larger universe observed over more CBAM events, including the definitive regime now in force, would lift power. None of these would overturn the present non-result; they would test whether a small, real effect lies beneath the resolution of this sample.

7. Conclusion

We asked whether the equity prices of global steel and aluminium producers moved in proportion to firms' carbon intensity around three 2023 milestones of the EU Carbon Border Adjustment Mechanism: the European Parliament's adoption of the regulation, its entry into force, and the start of the transitional reporting phase. Using one cumulative abnormal return per firm-event against a leave-one-out benchmark and a cross-sectional interacted event study with issuer-clustered inference, we do not detect a stable repricing. The focal carbon slope among non-EU firms is small and insignificant under the primary specifications (-0.9 percentage points per standard deviation, 95% CI $[-2.6, 0.8]$, at $[-1, +1]$), turns positive, with the rank-based and deletion estimates statistically significant, only after a single mis-scaled firm is neutralised, and the pooled slope is insignificant under CR2, wild-bootstrap, and randomization inference. The EU carbon slope is not separately identified in the raw standardised specification. The design is underpowered, with a minimum detectable effect of one to two and a half percentage points and equivalence bounds that cannot exclude one-point effects.

We therefore report insufficient evidence to detect a carbon-intensity association in this sample and design, not a clean or well-identified null. Nor do the data support the hypothesised negative association: after the invalid observation is down-weighted or removed, the significant estimates are positive, an opposite-direction finding that requires replication. For the period we study, the carbon-cost signal that CBAM introduces was not visibly impounded into the cross-section of exposed producers in proportion to the characteristic the policy targets, consistent with a transitional phase that carried no certificate cost before 2026 and with a carbon premium that is fragile even in broad cross-sections. Two measurement limitations, an EU-domicile proxy for exposure and emissions vintages disclosed after the events, and the limited power of a small cross-section, mean the design cannot support a stronger reading. Future work that builds a firm-level exposure measure, uses point-in-time emissions, substitutes an externally traded benchmark for the leave-one-out mean, and enlarges the firm universe and event set to include the definitive regime now in force would establish whether a small effect lies beneath the resolution of the present sample. The contribution is a documented non-detection on a timely, policy-relevant question, together with an EU-versus-non-EU metals panel on which that question can be revisited as CBAM operates.

Data availability

The curated daily price and firm-emissions panels, together with the analysis code that reproduces all tables and figures, are available from the author on request, subject to the original data-source licenses.

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